

PALETTE-Aug: A Simple Texture-Preserving Augmentation for Medical Imaging Domain Generalization

Anonymous CVPR submission

Paper ID *****

Abstract

Introduction. Deep learning for medical image segmentation generalizes poorly across imaging modalities, and annotating data for every new domain is prohibitively costly. Synthesis-based augmentation like SynthSeg address this by generating training images from label maps alone, but require dense label maps and discard the source image's texture. We introduce PALETTE-Aug, a texture-preserving alternative, and ask when texture matters for generalization. **Methods.** We benchmark PALETTE-Aug against five methods on eight segmentation tasks: five whose targets are bounded by an anatomical interface (healthy brain, multi-organ, healthy spine, mandible, pelvis) and three defined by tissue appearance (brain tumor, multiple sclerosis lesions, breast lesions). Each method trains on one modality and is evaluated in-domain and out-of-domain. An ablation isolates the effect of texture preservation. **Results.** PALETTE-Aug achieves the highest Dice on six of the eight tasks and significantly outperforms every competing method overall. In the ablation (six tasks), isolating texture preservation raises out-of-domain Dice by 15% and 16% relative on two interface-free targets (brain tumor, multiple sclerosis), versus at most 4% on interface-bounded ones (healthy brain: -3%); breast lesions, also interface-free, show no Dice gain. **Conclusion.** Texture preservation helps most where the target lacks an anatomical interface, and PALETTE-Aug generalizes best overall across eight segmentation tasks.

1. Introduction

Medical image segmentation models are trained on one acquisition setting and are frequently applied to another: a different scanner, sequence, or imaging modality than the one seen during training. MRI contrast is a protocol choice rather than a physical constant, so the same anatomy yields incompatible intensity distributions across sequences, and a network that has only seen T1-weighted brain images can fail outright on T2-weighted or FLAIR images of the same

anatomy [3, 34]. Acquiring and annotating training data for every contrast a model might encounter is not practical: manual segmentation is expensive, and clinical sites rarely share raw data across institutions [33, 51]. The single-source domain generalization setting — train on one labeled dataset in one contrast, generalize to unseen contrasts and modalities without any target-domain data — is therefore the setting that matters for deployment, and the one a growing body of work addresses [36, 45, 55, 57].

The dominant single-source strategy is label-generative domain randomization, popularized by SynthSeg [3]. Given a dense anatomical label map — a voxel-wise parcellation of the whole image, not the sparse structure-of-interest annotation a segmentation dataset normally provides — intensities are resampled from a per-label Gaussian mixture, producing images of arbitrary, unrealistic contrast that force the network to ignore contrast and rely on shape and layout [3]. This is effective, but carries two structural costs. First, it is *label-driven*: the dense parcellation it resamples from is a richer annotation than the task itself requires, so generating synthetic contrast costs exactly the expensive labeling the method aims to reduce reliance on. Second, and less discussed, it *does not preserve texture*: replacing every labeled region with piecewise Gaussian noise discards the real gradients, partial-volume boundaries, and unlabelled fine structure the acquired scan contains. Layout survives; texture does not.

We ask whether discarding texture is necessary for contrast robustness, or only convenient once a dense label map is assumed. Our central finding is that the answer depends on the *task*: preserving real intensity structure helps most where the target is defined by tissue appearance, and much less where an anatomical interface separates it from its surroundings. Prior work shows that texture carries diagnostic signal in such tasks: texture features separate MS lesions from normal-appearing white matter [56], and diffuse glioma infiltrates beyond every margin visible on conventional MRI [17, 47]. We show that this distinction determines where a texture-preserving augmentation helps substantially.

We exploit it with **PALETTE-Aug**, a single-source augmentation that needs no dense label map. **PALETTE-Aug** groups the foreground intensities of a scan into classes with 1-D k -means, splits each class into spatial sub-regions with a Voronoi partition, and applies a signed random affine remap to the *real* intensities of each sub-region. Because existing intensities are remapped rather than replaced by noise, edges, gradients and partial-volume detail survive inside each sub-region. Only the contrast between regions is randomized, including sign inversions that no acquisition protocol produces. An optional last step uses labels, and only the sparse task labels that any segmentation dataset already has.

We test the mechanism by intervention. Keeping the partition and the region target means, we replace only the real-intensity remap with SynthSeg-style noise fill, in fourteen training settings, and measure the change relative to the Dice the swap starts from. The largest relative effects are all on targets without a tissue interface (+44% and +18% on brain tumor, +39% on MS lesions, the last for all eight test patients), and no interface-bounded setting gains more than 6%. The converse does not hold: several appearance-defined settings, breast lesions among them, show no effect, so we state the result in one direction only (§4.3). A segmentation-free gradient measurement confirms that the swap changes texture as intended: the noise-fill rungs keep only the partition edges, and the real-fill swap restores the source texture.

We train one nnU-Net [22] configuration, varying only the epoch budget per dataset, and evaluate cross-contrast, cross-modality and cross-dataset transfer on eight tasks: healthy brain, MS lesion, brain tumor, abdominal organ, mandible, breast lesion, spine and pelvis. We compare against a no-augmentation baseline, two variants of SynthSeg (the released dense-label variant and our re-implementation of the unreleased EM variant), SRCSM [48], and Auglab [34], a published cross-modality augmentation pipeline. **PALETTE-Aug** is the best method overall: it has the best Dice on six of the eight tasks and is significantly better than every competitor across tasks, on both Dice and boundary distance. The best competitor switches between Auglab and SRCSM from task to task, while **PALETTE-Aug** stays at or near the top throughout.

In summary, we make the following contributions:

- **A task-level account of when texture preservation matters.** Preserving real intensity structure helps most, relative to the model’s starting point, where the segmentation target has no anatomical interface separating it from its surroundings. We test this with a controlled intervention rather than asserting it post hoc, and report where it does not extend.
- **A causal ablation isolating texture.** Swapping only the real-intensity remap for noise fill, with the partition fixed,

places the largest relative effects on no-interface targets in fourteen training settings. A segmentation-free gradient measurement confirms that the swap changes texture as intended.

- **PALETTE-Aug**, the single-source contrast augmentation that exploits this, requiring no dense anatomical label map and using only the sparse task labels a segmentation dataset already provides.
- **Breadth evidence.** Across cross-contrast, cross-modality and cross-dataset transfer on eight tasks and sixteen datasets, with one configuration reused throughout (Spine was run on a separate pipeline), **PALETTE-Aug** is best on six tasks and significantly better than every competitor across tasks. Where it does not win we report whether the method ahead is actually ahead: on healthy brain it is a tie, on mandible a real loss.

2. Related Work

2.1. Domain generalization for medical segmentation

Domain generalization (DG) targets unseen distributions without access to target data, unlike unsupervised domain adaptation, which assumes unlabelled target images and solves an easier problem [44]. Most medical DG methods are *multi-source*, interpolating across several labeled domains by meta-learning, feature alignment, frequency-space exchange, multi-frequency architectures, foundation-model scale, or diffusion-based stylization [2, 30, 33, 35, 42, 51] – all of which still rely on source diversity. We instead work in the *single-source* regime — one labeled dataset, no target data, no second source – the setting that matters when target-domain data is unavailable at training time, which is routine in clinical deployment [36, 57]. This rules out cross-domain interpolation and makes architecture-coupled methods orthogonal to our contribution: our mechanism acts at the input level and leaves the segmentation network a fixed nnU-Net [22].

2.2. Synthesis-based contrast-agnostic training

The dominant single-source strategy is label-generative domain randomization, introduced by SynthSeg [3]: intensities are sampled from a per-label Gaussian mixture over an anatomical segmentation, yielding images of arbitrary contrast that force contrast-invariance. This family carries two structural costs: it is *label-driven* (a dense anatomical label map is required per subject – SIAM shifts the dependence from label *quantity* to *quality* with curated templates, but each still needs a full dense parcellation [49]), and it is *texture-destroying*, since per-label Gaussian sampling replaces real gradients, partial-volume detail and unlabelled structure with piecewise noise, retaining layout but not texture. The lineage has been specialized to white-

matter and MS lesions and to atlas-based expectation-maximization [6, 7, 28]. Among SynthSeg’s own variants we take the original (*noEM*), which resamples intensities from a dense anatomical label map and never looks at the acquired scan, and the *EM* variant, which instead fits its per-class intensity model to the real image by expectation-maximization and so needs only the sparse task labels a segmentation dataset already provides, not a dense parcellation. The EM variant has no released implementation; ours follows the mechanism described in the original paper [3].

2.3. Texture as a segmentation signal

That local texture, not only intensity level, carries information a segmentation model can use is established independently of us: second-order gray-level statistics discriminate image content beyond simple intensity [18], and quantitative texture features from tumor regions carry prognostic signal across CT, MRI and PET [27]. Closer to our setting, Kang et al. disentangle structure from texture in cross-domain image translation with a texture co-occurrence loss, targeting the same failure mode we do — naive translation distorts fine texture and degrades downstream segmentation [24]; a texture-fidelity loss has likewise been added to GAN-based lesion augmentation [15]. That some targets resist boundary-based methods for lack of a sharp edge, and are better served by texture-based segmentation, has been argued directly for diffuse white-matter lesions [26]. None of this prior work, to our knowledge, connects texture preservation in an *augmentation* to *which* segmentation targets it helps; that connection, tested causally, is our contribution.

2.4. Single-source intensity and appearance augmentation

Where the first line of single-source methods synthesizes from a label map, a second manufactures appearance diversity directly from the real image, with no generative label model. General-purpose libraries stack intensity, spatial and artifact transforms on an nnU-Net trainer; the recent Auglab pipeline [22, 34] is one such GPU-accelerated framework and is the strongest competitor across our tasks, and BigAug stacks a comparable set of coarse intensity, quality and spatial perturbations under the same single-source DG protocol we use [57]. Our method inserts the PALETTE-Aug transform as one additional operator inside the Auglab pipeline, so the comparison against Auglab directly isolates PALETTE-Aug’s contribution. Closer to our operator are random-filter methods, which vary appearance with an untrained nonlinear filter: GIN applies randomly re-initialized convolutions for label-free appearance variants [36]; related variants establish the random-filter principle for natural images, perturb intensities per class using the task mask, add learned diffusion generators, or per-

turb the Fourier amplitude spectrum instead of the image itself [8, 31, 39, 45, 55, 58]. Our operator instead partitions the foreground into intensity classes and Voronoi sub-regions and applies a signed *affine* remap of the real intensities within each, invertible and gradient-preserving by construction. The closest peer is SRCSM [48]. Like PALETTE-Aug it perturbs region-wise in 3-D with one label-consuming step: it applies a fresh untrained convolution per anatomical class, smooths the resulting boundary artifacts, and histogram-matches test images to the source. The two differ in the transform: a random nonlinear convolution in SRCSM, an affine remap that keeps the relative intensity structure inside each region in PALETTE-Aug.

3. Method

We describe PALETTE-Aug (§3.1), how it is applied during training (§3.2), and the causal ablation used to isolate texture preservation as the source of its effect (§3.3).

3.1. The PALETTE-Aug transform

Let x denote the intensity volume of a training scan, min-max normalized to $[0, 1]$, and $F \subset \Omega$ its foreground voxels (an intensity threshold; $x_v > 0.01$ in our implementation). PALETTE-Aug applies four steps at every training iteration, illustrated end-to-end in Fig. 1 on one real CHAOS (abdominal MRI) and one real ON-Harmony (brain MRI) volume. Panels (b)–(d) show each step’s flat target values; only (e), the training input, shows the texture the transform produces.

1. Intensity partition. We run 1-D k -means on the foreground intensities $\{x_v : v \in F\}$ with k sampled per iteration from $\{2, \dots, 6\}$, producing k intensity classes $C_1, \dots, C_k$ (Fig. 1, panel (b)). This partition is unsupervised and requires no anatomical label map: it groups voxels by intensity alone, so the same operator applies unmodified to any input image regardless of its modality or contrast — CT, T1, T2, FLAIR, or otherwise. This step is occasionally skipped in favor of a single global remap (step 3 below, with $C = 1$); per-step probabilities are set once and never tuned per dataset, and are listed in Sec. 7.

2. Spatial sub-parcellation. Each intensity class C_i is further subdivided into spatial sub-regions by a Voronoi partition seeded with points sampled within C_i (Fig. 1, panel (c) shows the combined effect of this step and the remap of step 3), giving sub-regions $R_{i,1}, \dots, R_{i,m_i}$ that are contiguous in space as well as similar in intensity. This second partition breaks the spatial correlation of a purely intensity-based grouping: two voxels with the same intensity in different parts of the anatomy can land in different sub-regions

and receive independent contrast perturbations. Each class is usually, but not always, sub-split this way (Sec. 7).

3. Signed affine remap. For each sub-region R , let $\bar{x}_R$ denote its mean intensity, $\mu_R \sim U(0, 1)$ a resampled target mean, and draw $\alpha_R \sim \pm U(0.5, 2.0)$ (a magnitude in $[0.5, 2.0]$ with an independently sampled sign). Every voxel $v \in R$ is remapped as

$$y_v = \mu_R + \alpha_R (x_v - \bar{x}_R). \quad (1)$$

Because Eq. (1) is a per-region affine function of the *real* intensity x_v , it preserves the relative structure of the signal within each region: edges, gradients, and partial-volume transitions between adjacent voxels are scaled, not replaced. Only the region-to-region contrast relationship is randomized, and $\alpha_R < 0$ additionally allows local contrast inversions that no acquisition protocol produces on its own, widening the space of contrasts the network sees beyond what any single real sequence could supply. Background voxels are zeroed at this step (and remain zero through the rest of the pipeline). The result is occasionally Gaussian-blurred, mostly not (Sec. 7).

4. Anatomical-label affine remap (optional). Steps 1–3 are unsupervised, so the k -means/Voronoi partition need not respect a true anatomical boundary: when two adjacent tissues share similar intensity, step 3 can by chance assign them the same target contrast and erase that boundary rather than randomizing it. This step corrects for that: for each real anatomical label ℓ present, we resample a fresh (μ_ℓ, α_ℓ) and apply Eq. (1) *again*, restricted to ℓ ’s voxels, **on top of the step-3 output** rather than the original voxels (Fig. 1, panel (d)) – so it composes with, rather than replaces, the unsupervised remap, and ℓ ’s voxels still inherit the internal structure steps 1–3 imposed on them. Because label assignment is independent of the unsupervised partition, this reintroduces a full-strength contrast discontinuity at the true label border regardless of whether step 3 preserved it, sharpening the boundary cue available to the network – not the underlying anatomy. It consumes only the labels the segmentation task already requires, so it needs no additional annotation, and is optional: disabling it (`label_remap_prob=0`) degrades PALETTE-Aug gracefully to steps 1–3 alone, e.g. on an unlabelled dataset. A final foreground z-score normalization follows an optional second blur pass, giving the training input shown in panel (e) of Fig. 1. Per-step probabilities are set once, never tuned per dataset, and listed in full in Sec. 7.

3.2. Training protocol

PALETTE-Aug is applied on-the-fly as one contrast-augmentation operator inside an existing GPU-accelerated nnU-Net augmentation pipeline [22, 34], alongside that

pipeline’s standard spatial and artifact augmentations; it adds no learned parameters and no measurable training-time overhead. A single configuration – the same k -range, sub-region count, blur schedule, and α range (Sec. 7) – is tuned once and reused unchanged across every dataset and modality in Sec. 4; no dataset receives its own hyperparameter sweep.

3.3. Causal ablation: isolating texture preservation

PALETTE-Aug differs from SynthSeg in two ways at once: the partition (unsupervised k -means and Voronoi vs. an anatomical label map) and the fill (remapped real intensities vs. Gaussian noise). A direct comparison cannot separate them.

We construct a controlled variant, *noise-fill*, with the same k -means/Voronoi partition and the same random target mean μ_R for each sub-region as Eq. (1). Instead of remapping the real intensities, it fills each sub-region with SynthSeg-style Gaussian noise, $\mu_R + \sigma_R \epsilon$ with $\epsilon \sim \mathcal{N}(0, 1)$ and σ_R drawn uniformly from $[0.05, 0.25]$ per sub-region. The partition and the target means are PALETTE-Aug’s own, so a difference between the two isolates whether the real texture inside each region is kept or discarded. We report this comparison separately for targets with and without a tissue interface, assigned by anatomy (Sec. 4); the texture hypothesis predicts a larger relative gain on the latter.

4. Experiments

4.1. Setup

Protocol. Each model trains on one real source modality and is tested on every modality of the same task, the protocol Auglab uses for spine [34], here over eight tasks and with external cohorts wherever one exists. No target-domain data is seen in training. All methods share one nnU-Net [22] configuration and schedule and differ only in input augmentation; the epoch budget is fixed per dataset (Sec. 7). **Ours** is PALETTE-Aug added as one operator to the Auglab pipeline [34], so Ours and Auglab differ by exactly the PALETTE-Aug transform. Each method is trained on three train/validation folds, and all three models are scored on the same held-out test set. Significance tests are paired, use the patient as the unit, and are Holm-corrected [20] (Sec. 9).

Datasets. We evaluate on eight tasks drawn from sixteen datasets (full roster in Tab. 3): abdominal organs, healthy brain, brain tumor, MS lesions, mandible, breast lesions, spine and pelvic musculoskeletal anatomy, across MRI, CT and dental CBCT. Brain tumor and MS lesions use Brats2024-GLI [10] and Open-MS [29]. Every task trains on a single source modality at a time and is tested on every other contrast of that task *and* the training contrast it-

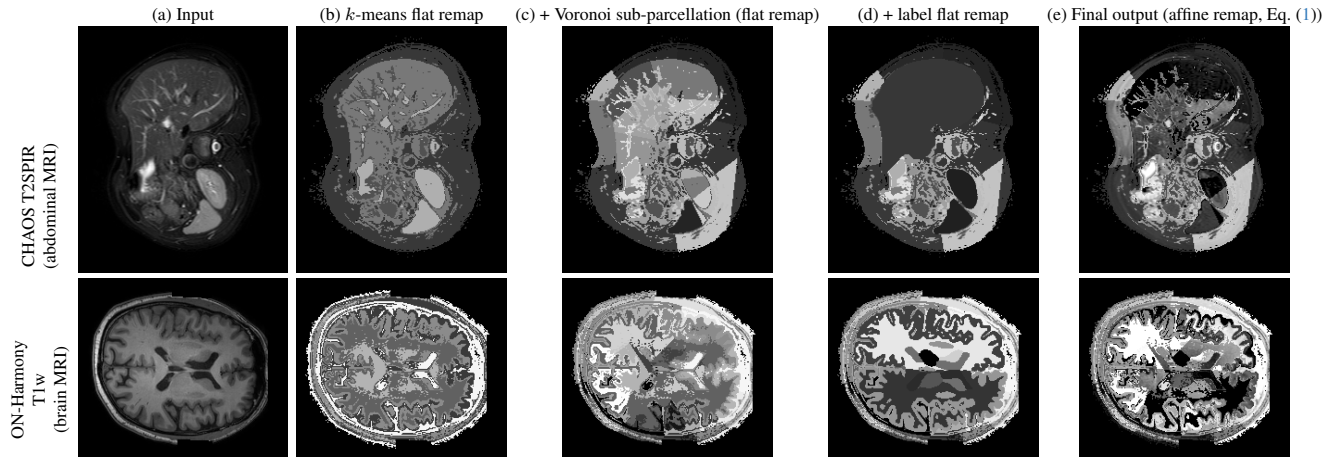


Figure 1. The PALETTE-Aug pipeline on one real CHAOS T2SPIR volume (top) and one real ON-Harmony T1w volume (bottom); every panel is a deterministic function of the same real volume (no external model), min-max normalized to $[0, 1]$ for display only. Panels (b)–(d) paint each region, sub-region, or label with its precomputed target mean μ from Eq. (1), isolating that flat value from the texture applied on top of it; panel (e) is the actual training input, produced by applying the full signed affine remap of Eq. (1) with real texture (not a flat fill) at both the region and, where available, anatomical-label level.

self — eighteen source-modality settings in total; no target modality is seen during training. Wherever an external cohort exists, its contrasts are pooled into that task as further held-out test columns rather than reported as a task of their own: CHAOS [25] transfers to CT from AMOS [23] and SLIVER07 [19]; Mandible (ToothFair2 [5], CBCT only) to the CT and MR of HaN-Seg [38] and the CT of PDDCA [40], so that, apart from its own CBCT test set, every held-out contrast of this task comes from another dataset; Breast (I-SPY2 [32]) to Duke-Breast-MRI [13, 41], I-SPY1 [21] and ACRIN-6698 [37], the last contributing diffusion-weighted MRI, a contrast neither training modality resembles; and Spine (a CT/Dixon cohort) to the T1w and T2w of SPIDER [50]. Pelvis is built from the TotalSegmentator CT and MRI releases [9, 53], two unpaired cohorts. ON-Harmony [52] reference labels are silver-standard FreeSurfer [12] parcellations of the T1w, registered to the other contrasts. External CHAOS volumes are cropped to CHAOS’s restricted field of view before prediction (Sec. 8).

Baselines. SynthSeg comes in two variants, which we refer to throughout by how they obtain their per-class intensity model: *noEM*, the released variant, which reads it off a dense anatomical label map; and *EM*, which estimates it from the image by expectation-maximization and therefore needs no dense parcellation. We compare against five methods. (1) *No-augmentation*, the nnU-Net default training recipe. (2) *SynthSeg-noEM* [3], the original label-generative variant (the implementation released by the authors), which synthesises each training image from a dense anatomical label map alone, ignoring the acquired scan entirely. (3)

SynthSeg-EM, the variant described in the original work but never released publicly, which we re-implement; unlike SynthSeg-noEM it does not require a dense anatomical label map, but it is not dense-label-free either — it is compatible with and relies on the sparse task labels naturally available in a segmentation dataset, which it uses to estimate its per-class intensity model. (4) *SRCSM*, a semantic random-convolution contrast-synthesis augmentation, included as a second strong synthesis-based competitor that is neither label-generative nor a member of the Auglab family. (5) *Auglab*, the published GPU-accelerated augmentation pipeline of Molinier et al. [34]. Auglab is a prior method in its own right, not an ablation of ours, and we report it among the competitors accordingly; our full method (Sec. 4.1) applies PALETTE-Aug *on top of* the Auglab pipeline, so the PALETTE-vs-Auglab comparison also functions as the add-one-component test isolating the contribution of the PALETTE-Aug transform.

4.2. Main comparison

PALETTE-Aug is the best method overall. It has the best Dice on six of the eight tasks. Across tasks it is significantly better than every competitor ($p < 10^{-10}$ each, Tab. 1). Within single tasks it significantly outperforms every competitor on five tasks (Brats-GLI, CHAOS, OpenMS, Breast, Spine), three of five on Mandible and Pelvis, and two of five on ON-Harmony. The cross-task test treats our eight tasks as fixed. Treating each task as one unit instead, which is the question of how a method fares on a new task, PALETTE-Aug beats Auglab on seven of eight tasks (one-sided sign test $p = 0.035$) and SRCSM on five of the seven tasks where SRCSM was run ($p = 0.23$).

	Brats-GLI	CHAOS	ON-Harm.	Open-MS	Mandible	Breast	Spine	Pelvis	p vs ours
Baseline	23.4	29.5	35.0	21.0	59.2	35.0	48.5	29.8	$< 10^{-10}$
SynthSeg-noEM	8.8	60.9	64.8	4.3	53.6	16.5	47.5	15.0	$< 10^{-10}$
SynthSeg-EM	43.1	84.0	66.3	33.9	76.3	39.6	80.6	62.5	$< 10^{-10}$
SRCSM	41.5	84.6	66.6	33.0	79.0	16.4	—	65.3	$< 10^{-10}$
Auglab	44.8	84.6	66.1	34.3	78.3	40.1	80.9	64.3	$< 10^{-10}$
PALETTE-Aug (ours)	45.9	85.9	66.5	38.5	77.7	41.2	82.3	65.6	—
competitors beaten ($p < 0.05$)	5/5	5/5	2/5	5/5	3/5	5/5	4/4	3/5	

Table 1. Task-level Dice (%). Each column pools that task’s held-out contrasts, all of its source modalities and, where one exists, its external cohorts, pooled by contrast group so that a modality tested by several external sources does not outvote one tested by fewer (Sec. 9). p : Holm-corrected one-sided sign-flip test on the macro-averaged per-patient difference, stratified by task with tasks weighted equally — a test *across* tasks, not an average *of* them; we deliberately print no cross-task mean Dice, since averaging a whole-liver Dice with a breast-lesion Dice yields a number no task’s clinician would recognize. The bottom row counts, per task, the competitors PALETTE-Aug beats with the same test run within that task alone. Spine has no SRCSM run (Tab. 3). HD95 is in Tab. 7.

SynthSeg-noEM samples intensities from a dense anatomical label map. It works only where such maps exist and falls below the no-augmentation baseline on six of the eight tasks. SynthSeg-EM, which estimates its intensity model from the task labels, closes most of that gap. SRCSM also falls below the baseline on Breast.

No competitor is reliably second. The best competitor alternates between Auglab (Brats-GLI, Open-MS, Breast, Spine) and SRCSM (ON-Harmony, Mandible, Pelvis), the two tie on CHAOS, and SRCSM fails on Breast (16.4 Dice). Picking the right one would require knowing the answer in advance. PALETTE-Aug is first or second on every task but one.

Where PALETTE-Aug does not win. The one-sided test above cannot show that a competitor is ahead, so on these tasks we also ran the reverse test on the same patient-level design (Holm-corrected across competitors within the task). On ON-Harmony SRCSM leads by 0.1 Dice and neither method is significantly ahead. On Mandible SRCSM (+1.3 Dice, $p = 6 \times 10^{-4}$) and Auglab (+0.6, $p = 2.5 \times 10^{-3}$) are significantly ahead, and PALETTE-Aug is third. In the Mandible ablation the texture step is significantly positive, and the drop coincides with adding the Auglab pipeline on top (Tab. 9).

Boundary distance. HD95 is reported in the supplement (Tab. 7). PALETTE-Aug has the lowest HD95 on six of the eight tasks (Brats-GLI, CHAOS, ON-Harmony, Open-MS, Spine, Pelvis) and is significantly better than every competitor across tasks. It ties Auglab on Breast and is significantly behind SRCSM, Auglab and the baseline on Mandible.

4.3. Causal ablation: when does texture matter?

PALETTE-Aug’s partition could help on its own, without the texture-preserving remap. We test this with a ladder in which each rung adds one ingredient: baseline $\rightarrow +k$ -means $\rightarrow +$ label remap $\rightarrow +$ Voronoi $\rightarrow$ real-intensity fill $\rightarrow +$ Auglab. The three partition rungs fill every region with label-style Gaussian noise, as SynthSeg does. All rungs run the same pipeline (Sec. 3) in the same order and only switch steps on. The step from +Voronoi to real fill changes only the fill: the partition, the number of sub-regions, the distribution of region target means and both blur passes are the same on each side. A change at that step is therefore due to preserving the real texture.

Fig. 2 and Tab. 2 support a graded result in one direction. We read the swap relative to the Dice it starts from: a few points on a nearly solved task are a boost, while the same points on a hard task are a large share of what the model can do. Measured this way, the largest gains are all on targets defined by tissue appearance: Brats-GLI T1c (+43.6%, +13.3 Dice) and T1n (+18.1%, +7.2), both significant after correction over the fourteen rows, and Open-MS FLAIR (+39.3%, +7.7), which improves for all eight test patients but cannot reach significance with eight. No interface row gains more than 6.2%. The largest of them, CHAOS T2spir, adds 5.1 points, but from 82.0% Dice, on organs the partition alone already segments well. Pooled per task (Fig. 2), the step is worth +14.8% on Brats-GLI and +15.8% on Open-MS, against +3.9% on CHAOS, +2.0% on Tooth-Fairy2 and −3.0% on ON-Harmony. The converse does not hold. Several appearance-defined settings show no Dice effect: Open-MS T1w, Brats-GLI T2w and FLAIR, and both Breast rows (−2.8% and +1.3%, four cohorts pooled by contrast). So a large relative texture gain indicates an appearance-defined target, but an appearance-defined target does not guarantee one.

ON-Harmony labels 31 brain regions in the FreeSurfer

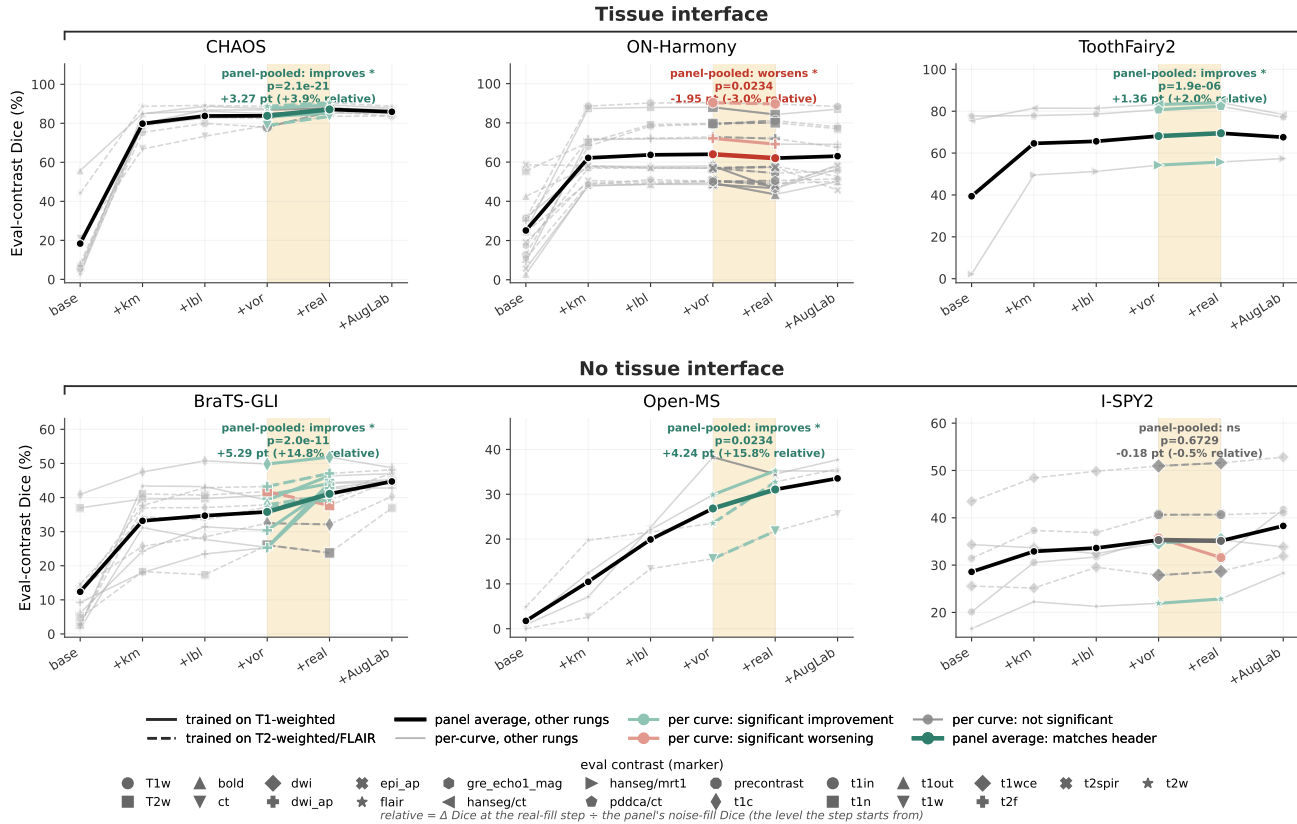


Figure 2. The one-variable test, one panel per task with a ladder, grouped by boundary type. Each thin line is one (training modality, held-out contrast) pair, external cohorts included, and only its real-fill segment (shaded band) is colored, by that pair’s own significance at this one step – teal = significant improvement, red = significant worsening, grey = not significant. The bold black line is the panel average; its real-fill segment is colored to match the panel-level test annotated above it, which pools all of the panel’s training modalities into one paired Wilcoxon test on *patients* (each counted once across contrasts and models), Holm-corrected across the six panels; the gain is also given relative to the panel’s noise-fill Dice. Two of the three no-tissue-interface panels improve significantly and strongly (Brats-GLI +14.8%, $p = 2.0 \times 10^{-11}$; Open-MS +15.8%, $p = 0.023$) and Breast does not move. The tissue-interface panels move much less in relative terms: CHAOS (+3.9%) and ToothFair2 (+2.0%) gain, and ON-Harmony worsens (−3.0%; significant with its eight sessions as units, which come from four subjects). Solid/dashed = trained on a T1-weighted vs. other contrast; marker = held-out contrast. Per-row numbers in Tab. 2; HD95 in Fig. 4 and Tab. 8.

label set [12]: ventricles, cortex, white matter, cerebellum, brainstem and eight deep grey nuclei. Every one is bounded by a tissue-type transition (CSF against tissue, grey against white matter), so we group it with CHAOS as an interface task. It is the only task where the swap hurts, perhaps because it labels nearly the whole brain; one densely labeled task cannot test this.

Does the fill swap change texture? Noise versus real fill. The ladder shows that the fill swap changes Dice, not that it changes texture. We check this without any segmentation, with the normalized-gradient-field (NGF) similarity [16] between each augmented volume and its source scan. NGF measures how well image gradients align: it is high when edges and fine structure are kept and falls to

about 1/3 for isotropic noise. Figure 3 follows NGF and Dice along the same rungs on Open-MS FLAIR. The three noise-fill rungs stay at NGF 0.45–0.46, above the noise floor because the partition edges survive but far below the source, while their Dice rises from 11.2 to 19.6. Those gains come from contrast randomization, not texture. Only the real-fill swap raises both, NGF to 0.81 and Dice to 27.3. On the same scale (Tab. 14, supplement), PALETTE-Aug (the real-fill rung, without Auglab) keeps significantly more source texture than Auglab (paired Wilcoxon over the 30 scans; FLAIR $p = 3.2 \times 10^{-3}$, T1w $p = 1.2 \times 10^{-6}$), and the label-style methods stay near the floor.

What the test does and does not show. It compares two versions of one method that differ in one component, so

Task	noise	Δ	rel. (%)	p
<i>Tissue interface</i>				
CHAOS T1in	85.7	+1.43	+1.7	4.0e-12
CHAOS T2spir	82.0	+5.11	+6.2	1.0e-20
ON-Harmony T1w [†]	63.2	-5.18	-8.2	0.070*
ON-Harmony T2w [†]	64.2	-1.55	-2.4	0.070*
ON-Harmony DWI [†]	64.6	+0.78	+1.2	0.094*
ToothFairy2 CBCT [‡]	68.1	+1.36	+2.0	4.9e-6
<i>No tissue interface</i>				
Open-MS FLAIR	19.6	+7.69	+39.3	0.070*
Open-MS T1w	34.1	+0.79	+2.3	1.0*
Brats-GLI T1n	39.8	+7.22	+18.1	2.8e-11
Brats-GLI T1c	30.5	+13.27	+43.6	4.3e-12
Brats-GLI T2w	41.0	+0.87	+2.1	0.38
Brats-GLI FLAIR	31.9	-0.19	-0.6	1.0
I-SPY2 T1-CE	30.8	-0.85	-2.8	0.26
I-SPY2 T2w	39.8	+0.50	+1.3	0.094

Table 2. Replacing label-style noise fill with the real-intensity remap, with the partition and region target means unchanged, for every training modality of every task with a ladder. noise: out-of-domain Dice with noise fill; Δ : change at the swap, in Dice points; rel.: Δ relative to the noise-fill Dice. p : two-sided paired Wilcoxon on patients, pooled over that row’s held-out contrasts, Holm-corrected across all fourteen rows. (Table 9 corrects within each ladder instead, a smaller family, so a few cells significant there are not significant here.) Every relative gain above 15% (bold) is on a no-tissue-interface target; no interface row exceeds 6.2%. *Open-MS has eight test patients and ON-Harmony eight test sessions (four subjects, each scanned at two sites), so no row of theirs can reach $p < 0.05$ after correction even when all eight move the same way, as they do for Open-MS FLAIR (8/8 improve) and ON-Harmony T1w and T2w (8/8 worsen). For ON-Harmony we treat the session as the unit; two sessions of one subject are not fully independent, so its p -values are, if anything, optimistic. [†]31 regions, each bounded by a genuine tissue-type transition (see text). [‡]Held-out axis is cross-dataset (HaN-Seg, PDDCA), since ToothFairy2 is CBCT only. HD95 in Tab. 8.

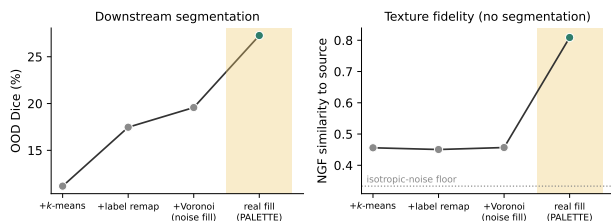


Figure 3. Noise versus real fill on Open-MS FLAIR. Left: out-of-domain Dice at each rung. Right: NGF similarity of the augmented volume to its source (mean $\pm$ s.e. over 30 scans), a segmentation-free measure of retained texture. The noise-fill rungs keep only the partition edges while Dice rises; the real-fill swap (shaded) lifts both.

differences in augmentation strength, partition granularity or training schedule cannot explain it. It is also separate from the benchmark ranking: on Mandible the texture step helps while the full method places third (§4.2).

Supporting analyses. The supplement adds a component ablation (Sec. 11), per-setting results, every ladder rung and the HD95 ablation (Sec. 9).

5. Discussion and Limitations

Why the effect follows the anatomy. Liver, spleen and kidney are encapsulated organs with a clear physical boundary, and edge-following methods have long segmented them well [25, 46]. An augmentation that scrambles their internal texture but keeps the boundary loses little. Diffuse glioma has no such boundary: tumor cells infiltrate beyond both the enhancing margin and the FLAIR abnormality [17, 47], so the target must be inferred from appearance. MS lesions are likewise defined by tissue signal [56]. This account interprets the intervention of Sec. 4.3; it is not separate evidence for it.

Appearance-defined is necessary, not sufficient. Open-MS T1w, Brats-GLI T2w and FLAIR, and both Breast rows show no texture effect. We have two untested hypotheses. The effect differs sharply by training contrast within one task (Brats-GLI: +13.3 from T1c, +7.2 from T1n, about zero from T2w and FLAIR), so how much discriminative texture the training contrast carries may matter as much as the target. And Breast pools four cohorts from different trials, which may dilute a cohort-specific effect. On interface tasks, where texture adds little, PALETTE-Aug’s gains likely come mostly from the wider range of region-to-region contrasts, including sign inversions, that PALETTE-Aug samples.

Limitations. We assign each target to the interface or no-interface group by its anatomy, not by a measured boundary statistic. We state the texture effect relative to the noise-fill Dice; in absolute points, CHAOS T2spir (+5.1) exceeds several appearance-defined settings. Open-MS has eight test patients and ON-Harmony eight sessions of four subjects, so their effects can be consistent without reaching significance, and ON-Harmony’s session-level tests are optimistic. Spine was run on a separate pipeline and has no SRCSM arm. The cross-task test treats our eight tasks as fixed; with the task as the unit, the advantage over SRCSM is not significant. PALETTE-Aug targets the single-source setting and is complementary to multi-source domain generalization, which assumes several labeled source domains.

6. Conclusion

We asked when preserving real image texture matters for contrast-robust segmentation. Its largest gains, relative to where the model starts, occur where the target is defined by tissue appearance rather than by an anatomical interface (+44% and +18% on brain tumor, +39% on MS lesions, at most +6% on interface targets), though not on every appearance-defined target. PALETTE-Aug, which preserves texture and needs no dense label map, is the best method overall across eight tasks and sixteen datasets. The task-level account makes a testable prediction for new tasks: a large relative gain from texture preservation should not be expected where the target has an anatomical interface.

References

- [1] Michela Antonelli, Annika Reinke, Spyridon Bakas, Keyvan Farahani, Annette Kopp-Schneider, Bennett A. Landman, Geert Litjens, Bjoern Menze, Olaf Ronneberger, Ronald M. Summers, et al. The medical segmentation decathlon. *Nature Communications*, 13(1):4128, 2022. 3
- [2] Jie Bao, Zhixin Zhou, Wen Jung Li, and Rui Luo. Structure-aware stylized image synthesis for robust medical image segmentation. *arXiv preprint arXiv:2412.04296*, 2024. 2
- [3] Benjamin Billot, Douglas N. Greve, Oula Puonti, Axel Thielscher, Koen Van Leemput, Bruce Fischl, Adrian V. Dalca, and Juan Eugenio Iglesias. SynthSeg: Segmentation of brain MRI scans of any contrast and resolution without retraining. *Medical Image Analysis*, 86:102789, 2023. 1, 2, 3, 5
- [4] Benjamin Billot, Colin Magdamo, You Cheng, Steven E. Arnold, Sudeshna Das, and Juan Eugenio Iglesias. Robust machine learning segmentation for large-scale analysis of heterogeneous clinical brain MRI datasets. *Proceedings of the National Academy of Sciences*, 120(9):e2216399120, 2023. 3
- [5] Federico Bolelli, Luca Lumetti, et al. Multi-structure segmentation in CBCT volumes: The ToothFairy2 challenge. *Medical Image Analysis*, 2026. 5, 2
- [6] Vicent Caselles-Ballester, Eloy Martínez-Heras, Giuseppe Pontillo, Zoe Mendelsohn, Elena M. Marrón, Juan Luis García Fernández, Laia Subirats, Jon Stutters, Jeremy Chataway, Frederik Barkhof, Sara Llufríu, and Ferran Prados. TimeLesSeg: Unified contrast-agnostic cross-sectional and longitudinal MS lesion segmentation via a stochastic generative model. *arXiv preprint arXiv:2605.07955*, 2026. 3
- [7] Stefano Cerri, Oula Puonti, Dominik S. Meier, Jens Wuerfel, Mark Muehlau, Hartwig R. Siebner, and Koen Van Leemput. A contrast-adaptive method for simultaneous whole-brain and lesion segmentation in multiple sclerosis. *NeuroImage*, 225:117471, 2021. 3
- [8] Boqi Chen, Yuanzhi Zhu, Yunke Ao, Sebastiano Caprara, Reto Sutter, Gunnar Rätsch, Ender Konukoglu, and Anna Susmelj. Generalizable single-source cross-modality medical image segmentation via invariant causal mechanisms. *arXiv preprint arXiv:2411.05223*, 2024. 3
- [9] Tugba Akinci D’Antonoli, Lucas K. Berger, Ashraya K. Indrakanti, et al. TotalSegmentator MRI: Robust sequence-independent segmentation of multiple anatomic structures in MRI. *Radiology*, 314(2), 2025. 5, 2
- [10] Maria Correia de Verdier, Rachit Saluja, Louis Gagnon, Dominic LaBella, Ujjwal Baid, Nourel Hoda Tahon, Martha Foltyn-Dumitru, Jikai Zhang, Ahmed W. Moawad, et al. The 2024 brain tumor segmentation (BraTS) challenge: Glioma segmentation on post-treatment MRI. *arXiv preprint arXiv:2405.18368*, 2024. 4, 2
- [11] Michael D. Ernst. Permutation methods: A basis for exact inference. *Statistical Science*, 19(4):676–685, 2004. 3
- [12] Bruce Fischl. FreeSurfer. *NeuroImage*, 62(2):774–781, 2012. 5, 7, 2
- [13] Lidia Garrucho, Kaisar Kushibar, Claire-Anne Reidel, Smriti Joshi, Richard Osuala, others, and Karim Lekadir. A large-scale multicenter breast cancer DCE-MRI benchmark dataset with expert segmentations. *Scientific Data*, 12(1):453, 2025. 5, 2
- [14] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673, 2020. 5
- [15] Qian Guan, Yuqing Chen, Zhun Wei, Ali Asghar Heidari, Huiling Hu, Xiao-Hong Yang, Jiao Zheng, Qiang Zhou, Huiling Chen, and Feng Chen. Medical image augmentation for lesion detection using a texture-constrained multichannel progressive GAN. *Computers in Biology and Medicine*, 145:105444, 2022. progressive GAN augmentation with an explicit texture-discrimination loss, added because standard GAN augmentation lost textural fidelity and hurt lesion detection. 3
- [16] Eldad Haber and Jan Modersitzki. Intensity gradient based registration and fusion of multi-modal images. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 726–733, 2006. 7
- [17] Alexander F. Haddad, Jacob S. Young, Ramin A. Morshed, and Mitchel S. Berger. FLAIRectomy: Resecting beyond the contrast margin for glioblastoma. *Brain Sciences*, 12(5):544, 2022. histopathological tumour infiltration inside peritumoral FLAIR abnormality, beyond the enhancing margin. 1, 8
- [18] Robert M. Haralick, K. Shanmugam, and Its’hak Dinstein. Textural features for image classification. *IEEE Transactions on Systems, Man, and Cybernetics*, SMC-3(6):610–621, 1973. foundational GLCM paper: second-order gray-level co-occurrence statistics carry discriminative content beyond intensity. 3
- [19] Tobias Heimann, Bram Van Ginneken, Martin A Styner, Yulia Arzhaeva, Volker Aurich, Christian Bauer, Andreas Beck, Christoph Becker, Reinhard Beichel, György Bekes, et al. Comparison and evaluation of methods for liver segmentation from CT datasets. *IEEE Transactions on Medical Imaging*, 28(8):1251–1265, 2009. 5, 2
- [20] Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979. 4, 3

- [21] Nola M. Hylton et al. Locally advanced breast cancer: MR imaging for prediction of response to neoadjuvant chemotherapy—results from ACRIN 6657/I-SPY TRIAL. *Radiology*, 263(3):663–672, 2012. 5, 2
- [22] Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl, Jens Petersen, and Klaus H. Maier-Hein. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18(2):203–211, 2021. 2, 3, 4, 1
- [23] Yuanfeng Ji, Haotian Bai, Chongjian Ge, Jie Yang, Ye Zhu, Ruimao Zhang, Zhen Li, Lingyan Zhanng, Wanwan Ma, Xiang Wan, et al. AMOS: A large-scale abdominal multi-organ benchmark for versatile medical image segmentation. In *Advances in Neural Information Processing Systems*, pages 36722–36732, 2022. 5, 2
- [24] Myeongkyun Kang, Philip Chikontwe, Dongkyu Won, Miguel Luna, and Sang Hyun Park. Structure-preserving image translation for multi-source medical image domain adaptation. *Pattern Recognition*, 144:109840, 2023. explicit structure-vs-texture disentanglement in cross-domain translation via a texture co-occurrence loss; motivated by naive translation distorting fine texture and degrading downstream segmentation – closest prior precedent for texture preservation as a design principle. 3
- [25] A. Emre Kavur, N. Sinem Gezer, Mustafa Barış, Sinem Aslan, Pierre-Henri Conze, Vladimir Groza, Duc Duy Pham, Soumick Chatterjee, Philipp Ernst, Savaş Özkan, et al. CHAOS challenge – combined (CT-MR) healthy abdominal organ segmentation. *Medical Image Analysis*, 69:101950, 2021. 5, 8, 2
- [26] Frithjof Kruggel, Jasmine S. Paul, and Hermann-Josef Gertz. Texture-based segmentation of diffuse lesions of the brain’s white matter. *NeuroImage*, 39(3):987–996, 2008. motivates texture-based over boundary/edge-based segmentation because diffuse WM lesions are inhomogeneous, unsharp and faint – implicit precedent for the texture-defined vs boundary-defined distinction. 3
- [27] Philippe Lambin, Emmanuel Rios-Velazquez, Ralph Leijenaa, Sara Carvalho, Ruben G. P. J. van Stiphout, Patrick Granton, Catharina M. L. Zegers, Robert Gillies, Ronald Boellard, Andre Dekker, and Hugo J. W. L. Aerts. Radiomics: Extracting more information from medical images using advanced feature analysis. *European Journal of Cancer*, 48(4):441–446, 2012. founding radiomics position paper; quantitative texture features carry prognostic signal across CT/MRI/PET and cancer types. 3
- [28] Pablo Laso, Stefano Cerri, Annabel Sorby-Adams, Jennifer Guo, Farrah Mateen, Philipp Goebel, Jiaming Wu, Peirong Liu, Hongwei Li, Sean I. Young, Benjamin Billot, Oula Puonti, et al. Quantifying white matter hyperintensity and brain volumes in heterogeneous clinical and low-field portable MRI. *arXiv preprint arXiv:2312.05119*, 2023. 3
- [29] Žiga Lesjak, Alfiia Galimzianova, Aleš Koren, Matjaž Lukač, Koen Van Leemput, and Žiga Špiclin. A novel public MR image dataset of multiple sclerosis patients with lesion segmentations based on multi-rater consensus. *Neuroinformatics*, 16(1):51–63, 2018. 4, 2
- [30] Da Li, Yongxin Yang, Yi-Zhe Song, and Timothy M. Hospedales. Learning to generalize: Meta-learning for domain generalization. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2018. 2
- [31] Heng Li, Haojin Li, Jianyu Chen, Mingyang Ou, Hai Shu, and Heng Miao. RaffesDG: Random frequency filtering enabled single-source domain generalization for medical image segmentation. *arXiv preprint arXiv:2405.01228*, 2024. 3
- [32] Wen Li, David C. Newitt, Jessica Gibbs, Lisa J. Wilmes, Ella F. Jones, Vignesh A. Arasu, and Nola M. Hylton. I-SPY 2 breast dynamic contrast enhanced MRI trial (ISPY2), 2022. Version 1, data set. 5, 2
- [33] Quande Liu, Cheng Chen, Jing Qin, Qi Dou, and Pheng-Ann Heng. FedDG: Federated domain generalization on medical image segmentation via episodic learning in continuous frequency space. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021. 1, 2
- [34] Nathan Molinier, Hendrik Möller, Thomas Dagonneau, Anna Curto-Vilalta, Robert Graf, Matan Atad, Daniel Rueckert, Jan S. Kirschke, and Julien Cohen-Adad. One sequence to segment them all: Efficient data augmentation for CT and MRI cross-domain 3D spine segmentation. *arXiv preprint arXiv:2605.03098*, 2026. 1, 2, 3, 4, 5
- [35] Ju-Hyeon Nam, Nur Suriza Syazwany, Su Jung Kim, and Sang-Chul Lee. Modality-agnostic domain generalizable medical image segmentation by multi-frequency in multi-scale attention. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 2
- [36] Cheng Ouyang, Chen Chen, Surui Li, Zeju Li, Chen Qin, Wenjia Bai, and Daniel Rueckert. Causality-inspired single-source domain generalization for medical image segmentation. *IEEE Trans. Med. Imaging*, 42(4):1095–1106, 2023. 1, 2, 3, 5
- [37] Savannah C. Partridge et al. Diffusion-weighted MRI findings predict pathologic response in neoadjuvant treatment of breast cancer: The ACRIN 6698 multicenter trial. *Radiology*, 289(3):618–627, 2018. 5, 2
- [38] Gasper Podobnik, Primoz Stojan, Primoz Peterlin, Bulat Ibragimov, and Tomaz Vrtovec. HaN-Seg: The head and neck organ-at-risk CT and MR segmentation dataset. *Medical Physics*, 50(3):1917–1927, 2023. 5, 2
- [39] Qiang Qiao, Wenyu Wang, Meixia Qu, Kun Su, Bin Jiang, and Qiang Guo. Medical image segmentation via single-source domain generalization with random amplitude spectrum synthesis. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2024. 3
- [40] Patrik F. Raudaschl et al. Evaluation of segmentation methods on head and neck CT: Auto-segmentation challenge 2015. *Medical Physics*, 44(5):2020–2036, 2017. 5, 2
- [41] Ashirbani Saha, Michael R. Harowicz, Lars J. Grimm, Connie E. Kim, Sujata V. Ghatge, Ruth Walsh, and Maciej A. Mazurowski. A machine learning approach to radiogenomics of breast cancer: a study of 922 subjects and 529 DCE-MRI features. *British Journal of Cancer*, 119(4):508–516, 2018. 5, 2

- [42] Sourya Sengupta, Satrajit Chakrabarty, Keerthi Sravan Ravi, Gopal Avinash, and Ravi Soni. SynthFM: Training modality-agnostic foundation models for medical image segmentation without real medical data. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. 2
- [43] Harshay Shah, Kaustav Tamuly, Aditi Raghunathan, Prateek Jain, and Praneeth Netrapalli. The pitfalls of simplicity bias in neural networks. In *Advances in Neural Information Processing Systems*, 2020. 5
- [44] Hyungseob Shin, Hyeongyu Kim, Sewon Kim, Yohan Jun, Taejoon Eo, and Dosik Hwang. SDC-UDA: Volumetric unsupervised domain adaptation framework for slice-direction continuous cross-modality medical image segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 2
- [45] Zixian Su, Kai Yao, Xi Yang, Qiufeng Wang, Jie Sun, and Kaizhu Huang. Rethinking data augmentation for single-source domain generalization in medical image segmentation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2023. 1, 3
- [46] Kenji Suzuki, Hieu Trung Huynh, Yipeng Liu, Dominic Calabrese, Karen Zhou, Aytekin Oto, and Masatoshi Hori. Computerized segmentation of liver in hepatic CT and MRI by means of level-set geodesic active contouring. In *Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pages 2984–2987, 2013. PMC4283827; edge-driven level-set/geodesic-active-contour segmentation of liver – cited as evidence that a gradient ridge exists at organ boundaries. Original entry mis-cited a different (2007, non-matching-title) author list under this PMC id; corrected 2026-08-03 after verification. 8
- [47] Jun Tang, Liangpeng Chen, Ziyang Li, Cong Li, and Deling Li. Imaging strategies for delineating glioma margins: Aiming at precise therapy. *Fundamental Research*, 6(4):2644–2653, 2025. "gliomas do not exhibit a clear boundary between malignant and healthy tissue"; infiltration several cm into normal-appearing parenchyma. Journal corrected 2026-08-03 (was mis-attributed to "Meta-Radiology", a different, unrelated journal). 1, 8
- [48] Franz Thaler, Martin Urschler, Mateusz Kozinski, Matthias A. F. Gsell, Gernot Plank, and Darko Stern. Semantic-aware random convolution and source matching for domain generalization in medical image segmentation. *IEEE Access*, 2025. 2, 3
- [49] Romain Valabregue, Ines Khemir, Eric Badinet, François Rousseau, Guillaume Auzias, and Reuben Dorent. SIAM: Head and brain MRI segmentation from few high-quality templates via synthetic training. *arXiv preprint arXiv:2605.02737*, 2026. 2
- [50] Jasper W. van der Graaf, Miranda L. van Hooff, Constantinus F. M. Buckens, Matthieu Rutten, Job L. C. van Susante, Robert Jan Kroeze, Marinus de Kleuver, Bram van Ginneken, and Nikolas Lessmann. Lumbar spine segmentation in MR images: a dataset and a public benchmark. *Scientific Data*, 11:264, 2024. 5, 2
- [51] Shujun Wang, Lequan Yu, Kang Li, Xin Yang, Chi-Wing Fu, and Pheng-Ann Heng. DoFE: Domain-oriented feature embedding for generalizable fundus image segmentation on unseen datasets. *IEEE Trans. Med. Imaging*, 39(12):4237–4248, 2020. 1, 2
- [52] Shaun Warrington, Andrea Torchi, Olivier Mougin, Jon Campbell, Asante Ntata, Martin Craig, Stephania Assimopoulos, Fidel Alfaro-Almagro, Karla L. Miller, Mark Jenkinson, Paul S. Morgan, and Stamatios N. Sotiropoulos. A multi-site, multi-modal travelling-heads resource for brain MRI harmonisation. *Scientific Data*, 12:595, 2025. 5, 1
- [53] Jakob Wasserthal et al. TotalSegmentator: Robust segmentation of 104 anatomic structures in CT images. *Radiology: Artificial Intelligence*, 5(5):e230024, 2023. 5, 2
- [54] Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945. 3
- [55] Zhenlin Xu, Deyi Liu, Junlin Yang, Colin Raffel, and Marc Niethammer. Robust and generalizable visual representation learning via random convolutions. In *International Conference on Learning Representations (ICLR)*, 2021. 1, 3
- [56] Jing Zhang, Longzheng Tong, Lei Wang, and Ning Li. Texture analysis of multiple sclerosis: a comparative study. *Magnetic Resonance Imaging*, 26(8):1160–1166, 2008. PMID 18513908; texture features discriminate MS lesion vs normal-appearing WM vs normal WM. 1, 8
- [57] Ling Zhang, Xiaosong Wang, Dong Yang, Thomas Sanford, Stephanie Harmon, Baris Turkbey, Bradford J. Wood, Holger Roth, Andriy Myronenko, Daguang Xu, and Ziyue Xu. Generalizing deep learning for medical image segmentation to unseen domains via deep stacked transformations. *IEEE Trans. Med. Imaging*, 39(7):2531–2540, 2020. 1, 2, 3
- [58] Haoyu Zhao, Wenhui Dong, Rui Yu, Zhou Zhao, Du Bo, and Yongchao Xu. MoreStyle: Relax low-frequency constraint of fourier-based image reconstruction in generalizable medical image segmentation. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2024. 3

PALETTE-Aug: A Simple Texture-Preserving Augmentation for Medical Imaging Domain Generalization

Supplementary Material

7. Training configuration

All methods share one nnU-Net [22] configuration; only the input augmentation differs. Nothing below is tuned per dataset.

Training schedule. 3D full-resolution nnU-Net, SGD with Nesterov momentum 0.99, initial learning rate 10^{-2} with the nnU-Net polyLR decay, 250 mini-batches per epoch, combined Dice and cross-entropy loss, deep supervision on. Patch size, spacing and batch size are whatever the nnU-Net planner selects for that dataset, identically for every method. The epoch budget is fixed per dataset and shared by all methods on it: 2500 (Brats-GLI), 2000 (ON-Harmony, Open-MS), 1000 (ToothFairy2, I-SPY2) and 200 (CHAOS, Pelvis); Spine was trained on a separate pipeline. Each method is trained as three models on three train/validation folds, all scored on the same held-out test set; folds are the same across methods.

PALETTE-Aug hyper-parameters. Set once and never tuned per dataset: k -means class count drawn per sample from $\{2, \dots, 6\}$, skipped in favor of a single global remap with probability 0.10; Voronoi sub-parcellation of each intensity class with probability 0.60 (else left as one region); signed affine coefficient $\alpha \in \pm U(0.5, 2.0)$ applied per sub-region as in Eq. (1) (always applied, to whatever partition results from the k -means/Voronoi steps); per-label affine decoupling applied independently per label with probability 0.5 (skipped for any label with under 4 foreground voxels in that sample); Gaussian blur σ drawn from $\{0, 0, 0, 0.3, 0.5, 0.8\}$, applied twice (after the region-level and after the label-level remap), so mostly not applied at either point. The transform is applied on the GPU, on-the-fly, to a fixed 50% of training iterations (*train050*); the remaining half of iterations use the un-transformed real image.

Validation-time configuration. *val000* (the configuration reported in the main paper) applies no synthetic contrast to the validation split, matching every competing method and keeping model selection like-for-like. *val100* applies PALETTE-Aug to the entire validation split; it is reported in Sec. 10.

Baseline configurations. SynthSeg-noEM uses the authors' released generator with its default priors. SynthSeg-EM is our re-implementation of the variant described but

not released in the original work; it estimates a per-class intensity model from the sparse task labels available in each dataset via expectation-maximization, and is otherwise identical in its sampling procedure. SRCSM and Auglab use their published default configurations. No competing method was re-tuned in our favor, and all use the same folds, schedule and epoch budget as above.

Compute. Training was performed on NVIDIA L40S and H100 GPUs. A single fold costs from under an hour (CHAOS, 200 epochs) to roughly three days (Brats-GLI, 2500 epochs); the full study is dominated by the six methods $\times$ eighteen settings $\times$ three folds of the main comparison, plus the fourteen ablation ladders of Sec. 4.3.

8. Datasets and evaluation protocol

All evaluation is performed on held-out test cases only. For each task the model is trained on a single source contrast and evaluated on every contrast of that task's test split, including the training contrast itself; training and validation subjects never appear in any reported number.

Task	Dataset	Train modalities	Test modalities
Multi-Organ	CHAOS	T1-in, T2-SPiR	T1-in, T1-out, T2-SPiR, CT
	AMOS	–	CT, MRI
	SLIVER07	–	CT
Brain Glioma	Brats2024-GLI	T1n, T1c, T2w, FLAIR	T1n, T1c, T2w, FLAIR
Brain MS	Open-MS	FLAIR, T1w	FLAIR, T1w, T2w
Healthy Brain	ON-Harmony	T1w, T2w, DWI	T1w, T2w, BOLD, DWI, EPI, GRE
Mandible	ToothFairy2	CBCT	CBCT
	HaN-Seg	–	CT, MR-T1
	PDDCA	–	CT
Breast	I-SPY2	T1-CE, T2w	T1-CE, T2w
	Duke-Breast-MRI	–	T1-CE, pre-contrast
	I-SPY1	–	T1-CE, pre-contrast
	ACRIN-6698	–	DWI
Spine [§]	CT/Dixon cohort	CT, Dixon in	CT, Dixon fat/in/water
	SPIDER	–	T1w, T2w
Pelvis	TotalSegmentator	CT, MRI	CT, MRI

Table 3. Per-task dataset roster. Rows marked “–” are cross-dataset transfer targets only, never trained on; their contrasts are extra held-out test columns of the task they sit under, not tasks of their own. Mandible’s non-CBCT test contrasts all come from other datasets, since ToothFairy2 acquires CBCT alone. Pelvis’s CT and MRI come from different patients. [§]Spine results were computed on a separate pipeline with the same method suite except SRCSM (Sec. 8). Counts are in Tab. 4.

ON-Harmony [52] — 31-region healthy-brain segmentation. Reference labels are *silver standard*: FreeSurfer par-

956	cellations [12] of each session’s T1w, grouped into 31 re-	Breast: I-SPY2, Duke-Breast-MRI, I-SPY1, ACRIN-	1005
957	gions and rigidly registered with ANTs into the native space	6698 [13, 21, 32, 37, 41] — breast-lesion segmentation	1006
958	of the same session’s other contrasts. They are not manual	on malignant-only cohorts. I-SPY2 (CC BY 4.0) trains, on	1007
959	annotations, so absolute Dice here is agreement with an au-	its first-post-contrast T1 (T1-CE) and its T2w; its reference	1008
960	tomated reference; every method is scored against the same	is the trial’s functional tumor volume mask. Three external	1009
961	reference. The test set is eight sessions of four subjects,	cohorts are test-only. Duke-Breast-MRI and I-SPY1 come	1010
962	each scanned at two sites.	through the MAMA-MIA release, whose expert primary-	1011
		lesion protocol both share, and contribute T1-CE and pre-	1012
963	Open-MS [29] — MS-lesion segmentation with manual	contrast columns; Duke-Breast-MRI is CC BY-NC 4.0, a	1013
964	consensus annotations, evaluated across FLAIR, T1w and	non-commercial term no other cohort here carries. ACRIN-	1014
965	T2w.	6698, the diffusion sub-study of I-SPY2 with no patient in	1015
		common with our I-SPY2 subjects, contributes the baseline	1016
966	Brats2024-GLI [10] — post-treatment glioma sub-	diffusion-weighted acquisition with the trial’s whole-tumor	1017
967	region segmentation (necrotic core, surrounding FLAIR hy-	DWI region, a contrast neither training modality resembles.	1018
968	perintensity, enhancing tumor, resection cavity), trained on	Because I-SPY2 is mostly acquired unilaterally, bilateral	1019
969	each of T1n, T1c, T2w and T2-FLAIR and evaluated on all	external volumes are cropped to the lesion side and, for	1020
970	four.	the full-thorax acquisitions, also anterior-posteriorly to the	1021
		breast-plus-chest-wall extent of I-SPY2’s own crops; with-	1022
971	CHAOS [25] — abdominal organ segmentation (liver,	out the second crop every method’s HD95 is dominated by	1023
972	spleen, both kidneys), evaluated on the CHAOS MR con-	the thorax. I-SPY2’s DCE series are acquired unilaterally	1024
973	trasts (four held-out MR patients) and CT (20 held-out pa-	at some sites and bilaterally at others; both field-of-view	1025
974	tients), and on two external cohorts: CT and T2w MRI from	variants are kept per patient, and the significance tests treat	1026
975	AMOS [23] and liver-only CT from SLIVER07 [19]. In	them as one patient.	1027
976	the pooled CT group, liver comes from CHAOS, AMOS		
977	and SLIVER07 and spleen and kidneys from CHAOS and	Spine [50] — vertebrae, intervertebral discs and spinal	1028
978	AMOS. CHAOS images a restricted axial field of view, so	canal. Results for this task were computed on a separate	1029
979	the full-torso external volumes are cropped to the CHAOS-	pipeline with the same method suite except SRCSM, trained	1030
980	equivalent axial slab (median CHAOS margins around the	on CT and Dixon in-phase MRI of a CT/Dixon cohort and	1031
981	kidneys for AMOS and the liver for SLIVER07) before pre-	tested on that cohort’s held-out CT and Dixon fat, in-phase	1032
982	diction; the network sees, and is scored on, only the cropped	and water images and on the T1w and T2w of SPIDER.	1033
983	volume.	Dice and HD95 are computed on each whole structure per	1034
		label (the pipeline’s <code>global_bin</code> metrics). No SRCSM	1035
984	ToothFairy2, HaN-Seg and PDDCA [5, 38, 40] —	arm was run, so Spine is absent from every SRCSM com-	1036
985	mandible segmentation, trained on dental cone-beam CT,	parison.	1037
986	the one non-CT, non-MRI training modality here. Tooth-		
987	Fairy2’s annotation coverage is cohort-dependent, so every	Pelvis [9, 53] — eleven pelvic and hip musculoskeletal	1038
988	column of this task is scored on the mandible alone (teeth	structures (both hips, sacrum, and the gluteus maximus,	1039
989	excluded, matching the external references), which keeps	medius and minimus and iliopsoas on each side), from the	1040
990	the numbers being averaged commensurable. ToothFairy2	TotalSegmentator CT (CC BY 4.0) and MRI (CC BY-NC-	1041
991	acquires CBCT only, so apart from its own 71 CBCT test	SA 2.0) releases. The two are unpaired cohorts, and the	1042
992	cases its held-out contrasts all come from other datasets:	MRI release mixes sequences by design, so its MRI column	1043
993	HaN-Seg CT and MR-T1 (42 cases each) and PDDCA CT	is a sequence-heterogeneous test rather than one named	1044
994	(40 cases), the two CT sources pooled as one contrast group.	contrast. CT is subsampled to 200 cases to keep training	1045
995	ToothFairy2’s volumes are stored superior-inferior inverted	time comparable with the other tasks.	1046
996	relative to the external cohorts — an inversion invisible to		
997	the image header, found by checking anatomy directly —	9. Per-setting results	1047
998	so the external test volumes are flipped into ToothFairy2’s		
999	convention before inference, which leaves Dice and HD95	Tab. 1 in the main paper reports results at the task level,	1048
1000	unchanged under the rigid transform. The correction mat-	pooling each task’s held-out test contrasts, all of its source	1049
1001	ters for the comparison: a method without flip augmentation	modalities and its external cohorts. This section reports	1050
1002	(SRCSM) was penalised by the inversion far more than the	the same evaluation broken out into the eighteen individ-	1051
1003	others, and it is after this correction that SRCSM leads this	ual source-modality settings. Every number is a cross-fold,	1052
1004	task.		

Task	Train	Test
CHAOS	16	24
ON-Harmony	84 ^a	8
Open-MS	22	8
Brats2024-GLI	630	70
ToothFairy2	408	71
I-SPY2	476	84
Pelvis (CT / MRI)	170 / 156	30 / 27

Table 4. Per-task subject counts for the training cohorts, pooled across the 3 folds used throughout (± 1 between folds). ON-Harmony counts are scan sessions; the eight test sessions are four subjects, each scanned at two sites. External cohorts are test-only and counted separately: AMOS and SLIVER07 (CHAOS), HaN-Seg and PDDCA (Mandible), Duke-Breast-MRI (291), I-SPY1 (167, after excluding three bilateral cancers and one implant by a rule fixed before prediction) and ACRIN-6698 (371) for Breast, and SPIDER (171 T1w, 219 T2w) for Spine, whose own CT/Dixon test split has 27–29 cases per contrast. I-SPY2 counts are patients; one held-out patient can yield more than one T2w image (both field-of-view variants). Pelvis MRI keeps 183 of the 298 released cases. ^a76 for the T2w-trained models.

cross-class average over all held-out contrasts of that setting, under the protocol of Sec. 4.1.

Statistical test construction. All tests are paired, and the unit of independence is the *patient*, never the fold, the image or the contrast. A patient’s score is its mean over labels and evaluated folds; where the same patient is scored in several held-out contrasts, or by the models of several training modalities, those scores move together as one unit under the null. Treating them as separate samples would be pseudo-replication and would inflate the effective sample size — a real risk here, since several tasks test a handful of patients under many contrasts (ON-Harmony: eight sessions of four subjects, six contrasts). For ON-Harmony the unit is the session, since its four subjects leave a test that could never reach significance; two sessions of one subject are not fully independent, so its p -values are, if anything, optimistic, and we do not rest any claim on ON-Harmony alone. Per-setting comparisons (below) use the same test within each setting; the ablation ladders use the paired Wilcoxon signed-rank test [54], since there each setting is judged on its own. All tests are Holm-corrected [20] for multiple comparisons.

Why the aggregate tests are sign-flip tests on a macro-average. The aggregate tests (Tab. 1 and the per-dataset headline blocks) are not paired Wilcoxon signed-rank tests, which is the more usual choice in this literature: SynthSeg+, from the lineage we compare against, marks a method as better than its competitors using a one-sided Bonferroni-corrected Wilcoxon signed-rank test on paired Dice [4]. We

therefore give our reason rather than leave it implicit. The *estimand* we test, by contrast, is not unusual at all — it is the same equal-weight-per-stratum principle the Medical Segmentation Decathlon adopted, for the same reason, as set out below.

The estimand. What we claim is $\text{macro}\Delta$: the mean, over held-out contrasts, of that contrast’s mean paired per-case difference (ours minus competitor) — equal weight per contrast, which is exactly what the *all* column of every table reports, so the test and the table measure the same thing. Equal weighting is not cosmetic here. The claim under test is cross-contrast generalization, and a contrast’s held-out case count reflects how large a public cohort happens to be, not how much that contrast should say about generalizing. CHAOS is the extreme case: its held-out CT evidence includes an external cohort contributing 200 cases, while each of its three MR contrasts contributes 8. Weighting by case count would let one CT cohort cast more than half the votes in a test whose conclusion is about transfer across MR contrasts.

This is the same problem, and the same answer, as the Medical Segmentation Decathlon’s. Facing sample sizes that “varied heavily across all tasks and target ROIs,” it ranked algorithms within each target ROI, averaged those ranks per task, and averaged the per-task ranks for the final score — choosing that over a flat average across ROIs precisely because the flat scheme means “tasks with multiple target ROIs ... would be over-weighted” [1]. Our $\text{macro}\Delta$ applies the same principle one level across, with the held-out contrast in place of the target ROI. Where we differ is only in how that weighted quantity is reached: the Decathlon arrives at it through counts of pairwise Wilcoxon outcomes at an uncorrected per-comparison α , whereas we test the weighted difference itself and Holm-correct across competitors.

Why not a pooled Wilcoxon. The signed-rank test is defined on a single undifferentiated sample of paired differences. Applying it to the concatenation of all contrasts therefore tests a case-weighted null — the imbalance above, reintroduced — and there is no standard reweighting of the signed-rank statistic that recovers the equal-per-contrast estimand. The disagreement this causes runs in both directions: a pooled test can declare a tenth of a Dice point significant because one large cohort supplies most of the cases, or miss a consistent effect confined to the smaller contrasts.

The test. Under the null that the two methods are exchangeable on a given patient, that patient’s paired differences are equally likely to carry either sign. Sign-flipping is therefore a valid randomization null for *any* statistic computed from those differences [11], $\text{macro}\Delta$ included, which is what makes it applicable where the signed-rank test is not. We report the one-sided p in the direction of the stated hypothesis, with the two-sided p retained as a secondary col-

umn, and give a 95% confidence interval from a bootstrap that resamples patients within groups of patients sharing the same set of contrasts, so each source keeps its observed size.

Evaluating the tail. The statistic is a weighted sum of per-patient contributions, so its sign-flip null distribution can be computed rather than sampled. We enumerate all 2^M sign patterns of the $M=20$ patients with the largest contributions exactly, and treat the remaining, lighter patients as a Gaussian term with their exact null variance; with twenty patients or fewer the test is therefore exact. A plain normal approximation to the whole null, which an earlier version of this analysis used, is badly conservative in exactly the designs we have, where a few patients spanning several contrasts carry most of the null variance: on the CHAOS table it returned $p = 0.013$ for a comparison that a 200,000-draw Monte-Carlo test places below 5×10^{-6} . The enumerated tail is deterministic and reaches far tails without the $1/B$ floor of resampling. Its corollary is a hard floor for small cohorts: eight patients admit $2^8 = 256$ sign patterns, so no one-sided p can fall below $1/256$ however consistent the effect.

What is reported alongside. Every generated significance report also carries a per-contrast table of mean Δ with the raw, uncorrected Wilcoxon p for each contrast individually, and the ablation ladders use the paired Wilcoxon throughout. The standard test is reported; it is simply not the one the pooled headline numbers rest on.

Contrast-group pooling. For CHAOS, the same tested modality is sometimes evidenced by more than one source: CHAOS's CT evidence comes from its own held-out CT split plus AMOS and SLIVER07. Averaging every raw column with equal weight – the natural first implementation – silently lets a modality with more contributing sources outvote one with fewer in both the displayed *all* average and the significance test, since it then supplies more of the equally-weighted columns being averaged. We instead pool disjoint sources of the *same* modality by concatenating their cases into one group before any averaging happens, then average modality groups with equal weight – for CHAOS this additionally means averaging equal-weight across liver/spleen/kidney *within* the pooled CT evidence, since AMOS's own CT column already blends all three organs into one number before it reaches this step. The same pooling applies to Mandible (HaN-Seg and PDDCA both supply CT) and Breast (I-SPY2, Duke-Breast-MRI and I-SPY1 supply T1-CE, and Duke-Breast-MRI and I-SPY1 also supply pre-contrast); the remaining tasks have no contrast evidenced by more than one source.

Tables 5 and 6 break each task column of Tab. 1 into its training modalities. Where a margin is not significant in a single setting, the honest reading is that one setting cannot separate the methods, not that they are equivalent: sev-

Setting	Base	noEM	EM	SRCSM	Auglab	Ours
Brats-GLI T1n	25.3*	8.7*	43.1*	40.3*	46.3	46.4
Brats-GLI T1c	28.3*	9.5*	42.9*	44.5*	46.6	47.1
Brats-GLI T2w	18.1*	8.4*	44.1*	43.8*	47.1	47.3
Brats-GLI FLAIR	21.8*	8.7*	42.3	37.5*	39.4*	43.0
CHAOS T1in	29.5*	63.3*	84.4*	84.5	84.7*	86.0
CHAOS T2spir	29.6*	58.5*	83.5*	84.8*	84.5*	85.8
ON-Harmony T1w	27.5*	66.0*	67.3*	66.4	67.6*	68.6
ON-Harmony T2w	34.8*	66.7	67.5	66.7	67.2	67.7
ON-Harmony DWI	42.6*	61.6	64.2	66.8	63.4	63.1
Open-MS FLAIR	25.2*	4.6*	32.8*	34.6*	32.8*	39.5
Open-MS T1w	16.7*	4.0*	35.1*	31.4*	35.8	37.4
Mandible CBCT	59.2*	53.6*	76.3*	79.0	78.3	77.7
Breast T1-CE	34.1*	16.6*	40.4*	18.0*	41.1*	41.9
Breast T2w	35.9*	16.5*	38.9*	14.8*	39.0*	40.5
Spine CT	27.0*	46.7*	79.1*	—	77.5*	80.2
Spine Dixon in	69.9*	48.4*	82.2*	—	84.2	84.3
Pelvis CT	35.6*	17.1*	66.5*	71.4	68.5	69.2
Pelvis MRI	24.0*	12.8*	58.6*	59.2*	60.0	62.0

Table 5. Per-setting Dice (%), $\uparrow$, best per row in bold. Each row is one training modality of one task, scored on exactly the held-out contrasts and external cohorts of its task column in Tab. 1, so the rows of a task average to that column. *: PALETTE-Aug significantly better than that method within the setting (same patient-level sign-flip test, Holm-corrected across competitors). Spine has no SRCSM run.

Setting	Base	noEM	EM	SRCSM	Auglab	Ours
Brats-GLI T1n	41.0*	78.0*	18.4*	20.8*	15.1	15.4
Brats-GLI T1c	40.3*	78.3*	19.5*	14.9	15.1	15.4
Brats-GLI T2w	54.9*	79.7*	16.5*	15.7*	14.8	15.0
Brats-GLI FLAIR	43.1*	76.0*	17.3	18.5*	19.2*	16.6
CHAOS T1in	119.0*	78.0*	25.4	22.7	30.4*	22.3
CHAOS T2spir	125.1*	83.2*	22.0	26.0	25.1	23.2
ON-Harmony T1w	33.4*	10.4*	5.4	6.7	5.2*	4.3
ON-Harmony T2w	23.7*	4.6*	4.0	4.6*	4.2*	4.0
ON-Harmony DWI	14.7*	4.9	4.6	4.5	4.6	4.7
Open-MS FLAIR	39.9*	41.5*	23.6	23.8	26.6	24.3
Open-MS T1w	30.5*	42.3*	25.9	25.5	26.3*	24.2
Mandible CBCT	21.7	32.9*	24.9*	19.3	22.1	23.5
Breast T1-CE	61.8*	103.2*	61.7*	92.4*	57.3	59.0
Breast T2w	55.2	99.1*	60.1*	91.2*	56.9	56.3
Spine CT	78.3*	110.4*	4.5*	—	5.8*	3.9
Spine Dixon in	21.6*	113.2*	3.5*	—	3.6	3.1
Pelvis CT	146.2*	252.0*	20.0	16.0	23.5	21.6
Pelvis MRI	152.3*	271.6*	48.0	51.9*	47.8*	36.4

Table 6. Per-setting HD95 (mm), $\downarrow$, best per row in bold. Each row is one training modality of one task, scored on exactly the held-out contrasts and external cohorts of its task column in Tab. 1, so the rows of a task average to that column. *: PALETTE-Aug significantly better than that method within the setting (same patient-level sign-flip test, Holm-corrected across competitors). Spine has no SRCSM run.

eral settings have few test units (eight patients on Open-MS, eight sessions on ON-Harmony), which is why the main

1186

1187

claims rest on the task- and cross-task-level tests.

Boundary distance at task level. Tab. 7 gives the HD95 counterpart of Tab. 1. Across tasks PALETTE-Aug is significantly better than every competitor, Auglab included. Per task it has the lowest HD95 on Brats-GLI, CHAOS, ON-Harmony, Open-MS, Spine and Pelvis (on CHAOS the gap to SRCSM is not significant); it ties Auglab on Breast without a significant gap; and on Mandible it is significantly behind SRCSM, Auglab and the no-augmentation baseline, the one task where the baseline beats PALETTE-Aug. When HD95 disagrees with Dice for a method – a noticeably better HD95 than its Dice would predict – the usual explanation is low recall: a method that predicts only a small, confident, well-localized fragment of the structure can have a deceptively good boundary distance while missing most of the target, so a strong HD95 paired with a weak Dice is a warning sign rather than evidence of competitiveness.

Full ladder: numbers and significance. Fig. 2 (main paper) shows the rung sequence visually; Tab. 9 gives the underlying out-of-domain Dice at every rung of all fourteen ladders, with significance for each step against the rung immediately before it (Holm-corrected paired Wilcoxon on patients, within each ladder’s family of steps). Most steps that add an ingredient help, as expected. Eight cells are significant *decreases* (marked ↓). Two are the fill-swap step on ON-Harmony T1w and T2w. Two are the Voronoi sub-parcellation step, on Brats-GLI T1n and CHAOS T1in, which the real-fill step then more than recovers. The other four are the +Auglab rung — on ON-Harmony DWI, on CHAOS T1in, and twice on Mandible, the task where the complete method places third (Sec. 4.2). There PALETTE-Aug alone (the +real rung, 69.5) is the ladder’s maximum and adding the Auglab recipe takes it back down, which places the Mandible shortfall in the wrapper rather than in the transform this paper argues for. Both CHAOS ladders peak at the same rung. We have not diagnosed which Auglab operator is responsible.

We likewise do not hide that Voronoi’s own marginal contribution is task-dependent rather than uniformly positive – a separate paired significance check (same protocol as Tab. 13’s all-contrast Dice, not just the OOD slice used above) confirms the same pattern: Voronoi significantly *helps* Open-MS FLAIR and (by a small margin) ON-Harmony T1w, is flat on CHAOS T1in, and significantly *hurts* Brats-GLI T1n under the real-fill deployed configuration as well, not only here under noise fill. The full method still wins on Brats-GLI T1n regardless (Tab. 1): whatever Voronoi costs there is more than recovered by the subsequent real-intensity fill step, so the net effect of the complete pipeline is positive even though one internal component is

not.

Per-label breakdown: is the fill-swap effect uniform within a task? Fig. 2 shows that Brats-GLI T2w’s pooled fill-swap effect is small and not significant, with one significantly negative held-out contrast inside it (T2w→T1n, −4.1 Dice). This raises a further question: within that one (training, eval) pair, does the fill-swap step affect every anatomical sub-region equally, or is the worsening itself concentrated in specific labels? We break Brats-GLI T2w→T1n down by its four labels – necrotic core (NCR), oedema (SNFH), enhancing tumor (ET), resection cavity (RC); $n = 70$ test cases for SNFH and ET, present in essentially every case (NCR $n = 51$, RC $n = 67$, since not every tumor has a necrotic core or a resection cavity).

The worsening is concentrated in SNFH (oedema). SNFH has the largest gap between directions (+19.9 Dice) and is the only label whose fill-swap effect reverses sign significantly: +14.7 when trained on T1n, −8.8 when trained on T2w. RC shows the same reversal at a smaller size. NCR shows no significant effect in either direction, and ET improves only in the T1n→T2w direction. Fig. 5 shows this on one case: with real fill, the T2w-trained model’s SNFH prediction shrinks to a thin band near the ventricle, while its ET prediction is almost unchanged.

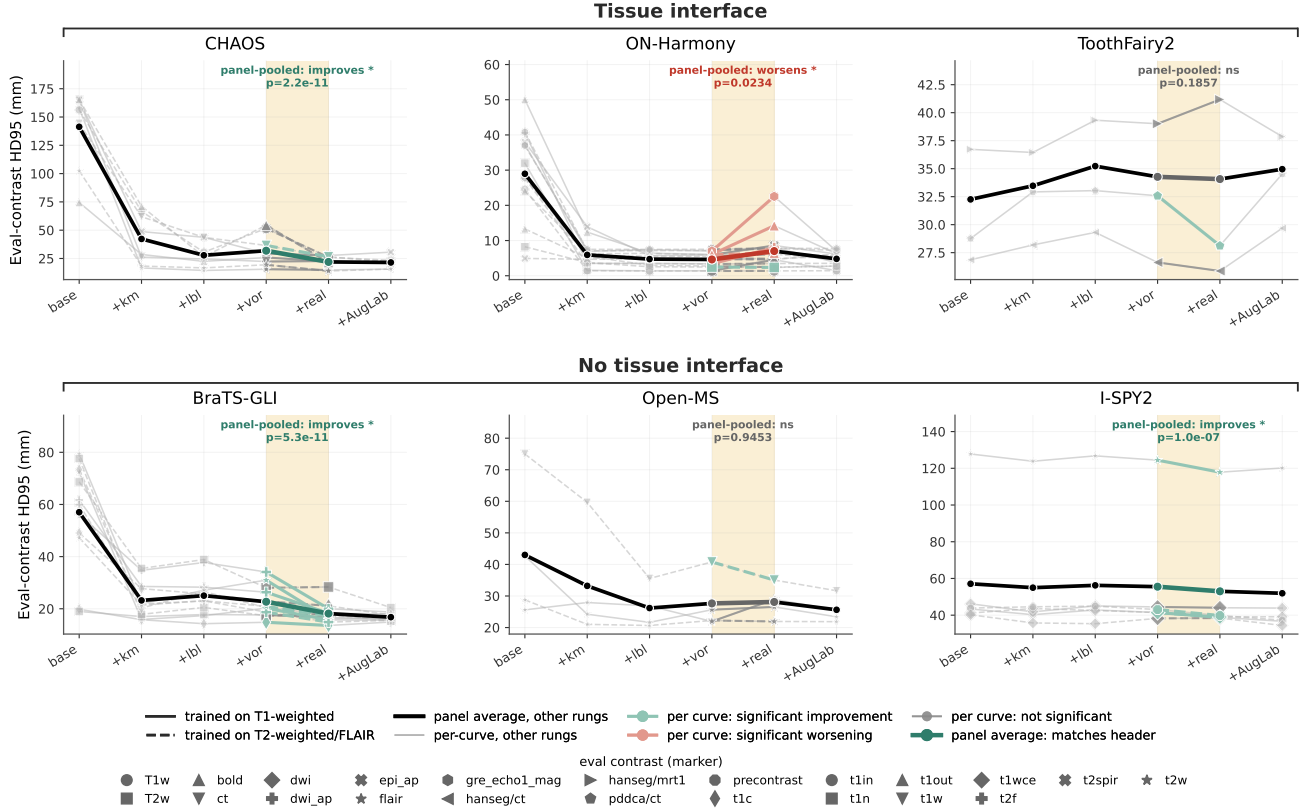
One candidate mechanism, flagged for future work rather than tested here: a texture pattern learned from one contrast’s rendering of a diffuse, texture-defined label like oedema may not simply fail to transfer to a different contrast’s rendering of the same tissue – it may actively mislead, if the model has come to rely on a contrast-specific cue that the other contrast does not reproduce. This would be an instance of shortcut learning [14] – specifically, a network relying on a domain-specific cue that does not reproduce across acquisition protocols, the mechanism Ouyang et al. [36] identify behind failed single-source domain generalization in cross-modality and cross-sequence medical segmentation – rather than the intended, causal feature, which a simplicity-biased network need not prefer even when it would generalize better [43]. We have not tested this mechanistically here; it is the most concrete hypothesis consistent with the label-specific, direction-dependent pattern above, not a demonstrated cause.

10. Synthetic contrast at validation time

Throughout the main paper PALETTE-Aug is applied with synthetic contrast during training only (*train050/val000*), because every competing method is likewise evaluated with a real-contrast validation set; this keeps the comparison exactly like-for-like. PALETTE-Aug can additionally be applied to the validation set, so that model selection is performed under randomized contrast as well (*train050/val100*).

	Brats-GLI	CHAOS	ON-Harm.	Open-MS	Mandible	Breast	Spine	Pelvis	p vs ours
Baseline	44.8	122.1	24.0	35.2	21.7	58.5	49.9	149.3	4.7e-48
SynthSeg-noEM	78.0	80.6	6.7	41.9	32.9	101.1	111.8	261.8	2.9e-58
SynthSeg-EM	17.9	23.7	4.7	24.7	24.9	60.9	4.0	34.0	1.8e-6
SRCSM	17.5	24.4	5.3	24.6	19.3	91.8	—	33.9	2.0e-34
Auglab	16.1	27.8	4.7	26.5	22.1	57.1	4.7	35.6	1.0e-4
PALETTE-Aug (ours)	15.6	22.7	4.3	24.3	23.5	57.7	3.5	29.0	—

Table 7. Task-level HD95 (mm, ↓). Same pooling and test as Tab. 1, and likewise no cross-task average. Spine has no SRCSM run.

Figure 4. The causal-ablation ladder of Fig. 2 (main paper), same per-dataset panels, same boundary-type grouping, same fill-swap rung and same visual scheme (per-curve color only on the real-fill segment; bold black panel average; one panel-pooled p -value per panel), reported on HD95 instead of Dice (*improvement* means the fill-swap step lowers HD95). Brats-GLI, CHAOS and Breast improve significantly at panel level and ON-Harmony worsens; Open-MS and ToothFairy2 do not move significantly. Underlying per-row deltas in Tab. 8.

Validation-time randomization improves most tasks on both metrics (Tab. 12), most visibly on boundary distance for CHAOS (22.7 $\rightarrow$ 20.6 mm). The effect is not significant across tasks, and Open-MS Dice moves slightly the other way, so we do not claim it as a separate contribution; we report it because it costs nothing and because it indicates that the benefit of contrast randomization extends to model selection and is not confined to the gradient updates.

11. Component ablation

To attribute the gain of the full method to its individual components, rather than to synthetic-contrast augmentation in general, we build an additive ablation ladder on top of the no-augmentation anchor (*Base*, Sec. 4.2): first the unsupervised k -means intensity partition alone (§3.1, step 1); then the anatomical label-remap step (step 4); then the Voronoi spatial sub-parcellation (step 2); and finally the full signed affine remap of Eq. (1) (step 3), which together constitute PALETTE-Aug on its own. Every rung is evaluated without

Task	boundary type	Δ HD95 (mm)	p
CHAOS T1in	interface	-1.36	0.80
CHAOS T2spir	interface	-18.47	3.4e-12
ON-Harmony T1w	interface	+6.82	0.062
ON-Harmony T2w	interface	+0.38	0.062
ON-Harmony DWI	interface	-0.05	0.75
ToothFairy2 CBCT	interface	-0.20	0.37
Open-MS FLAIR	no interface	-3.02	0.14
Open-MS T1w	no interface	+3.94	0.80
Brats-GLI T1n	no interface	-4.59	6.3e-10
Brats-GLI T1c	no interface	-9.91	4.2e-11
Brats-GLI T2w	no interface	-1.95	0.013
Brats-GLI FLAIR	no interface	-1.71	0.21
I-SPY2 T1-CE	no interface	-3.21	8.8e-5
I-SPY2 T2w	no interface	-1.82	0.0024

Table 8. Same fill-swap step as Tab. 2 (main paper), on HD95 (negative is better), with p Holm-corrected across the fourteen rows. Five of the six significant improvements are on no-tissue-interface targets, including both Breast rows, whose Dice does not move; the sixth, and the largest in millimetres, is CHAOS T2spir. HD95 is less consistent than Dice within a task (its sign flips between the two modalities of Open-MS), so we state the claim on Dice and report HD95 here rather than as confirmation of it.

Rung	Open-MS FLAIR	Open-MS T1w	Brats-GLI T1n	Brats-GLI T1c	Brats-GLI T2w	Brats-GLI FLAIR	CHAOS T1in
base	2.4	1.1	16.5	15.9	6.0	11.0	17.0
+km	11.2*	9.8*	38.4***	29.6***	38.6***	26.1***	82.2***
+lbl	17.5*	22.4*	41.8***	30.3	40.2**	26.4	86.1***
+vor	19.6	34.1*	39.8***↓	30.5	41.0	31.9***	85.7***↓
+real	27.3*	34.9	47.0***	43.7***	41.8	31.7	87.1***
+AL(v000)	30.6	36.4	46.9	44.2	46.8***	40.9***	86.4***↓
+AL(v100)	31.3	34.7	47.6*	46.5***	46.9	41.2	86.1

Rung	CHAOS T2spir	ON-Harm. T1w	ON-Harm. T2w	ON-Harm. DWI	Mandible CBCT	Breast T1-CE	Breast T2w
base	19.7	14.8	24.5	36.1	39.4	23.7	33.5
+km	77.4***	62.5*	63.5*	60.4*	64.6***	28.8***	37.0***
+lbl	81.4***	62.9	63.6	64.5*	65.6***	28.5	38.8***
+vor	82.0***	63.2	64.2*	64.6	68.1***	30.8***	39.8**
+real	87.1***	58.0*↓	62.6*↓	65.4*	69.5***	29.9	40.3*
+AL(v000)	85.4	64.2	63.9	60.9*↓	67.6***↓	34.7***	41.9***
+AL(v100)	85.9	64.2	64.4*	60.9	67.5***↓	—	—

Table 9. Out-of-domain Dice (%) at every rung of every ladder (Fig. 2), with significance for each step against the rung above it: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (Holm-corrected paired Wilcoxon on patients, within each ladder); ↓ marks a significant decrease. Rungs: base(line), + k -means, +label-remap, +Voronoi (noise fill), +real-intensity fill (PALETTE-Aug alone), +Auglab (val000/val100). Open-MS and ON-Harmony have eight test units each, so their smallest attainable p after correction is just under 0.05. I-SPY2 has no val100 arm. Mandible’s held-out axis is cross-dataset.

Auglab, so that the ladder isolates PALETTE-Aug’s own internal components from the separate question of whether it composes with a prior augmentation pipeline, which we address next. The synthetic-data fraction is fixed at 50% of training iterations throughout, as one of several augmenta-

Label	T1n→T2w	T2w→T1n	gap
NCR	21.5	20.0	+1.4
SNFH	66.3	46.4	+19.9
ET	27.5	29.0	-1.5
RC	52.3	42.8	+9.5

Table 10. Absolute OOD Dice (%) of the real-fill rung (PALETTE-Aug alone), per label, both transfer directions.

Label	T1n→T2w Δ	T2w→T1n Δ
NCR	+1.9 ($p=0.10$)	+1.0 ($p=0.77$)
SNFH	+14.7 ($p=4.3e-11$)	-8.8 ($p=1e-4$)
ET	+2.3 ($p=0.017$)	+1.6 ($p=0.24$)
RC	+2.9 ($p=0.0003$)	-6.9 ($p=0.0004$)

Table 11. Fill-swap step (+Voronoi noise fill → real fill), per label, both directions. p : Holm-corrected paired Wilcoxon, family = the 4 labels within each direction. **Bold** = significant.

BraTSGLI02108101, fold 0 → T2w-trained model evaluated on T1n

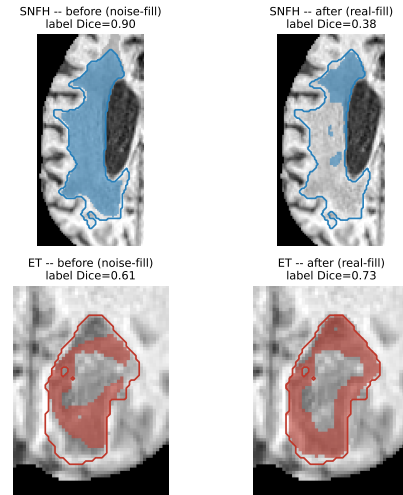


Figure 5. BraTSGLI02108101, fold 0, T2w-trained model evaluated on T1n. Zoomed crops before (noise-fill) and after (real-fill) the fill-swap step, for the SNFH-dominant (top) and ET-dominant (bottom) lesion components of the same case. Color = ground truth wherever the prediction agrees; grey = false positive; outline = ground truth.

tion constants (alongside, e.g., the k -means class count and the affine magnitude range of §3.1) that are set once and not tuned per dataset; validation uses real-contrast data only (no synthetic contrast at validation time).

Composability with Auglab. PALETTE-Aug is deployed, throughout this paper, as one additional operator inside the Auglab pipeline rather than as a replacement for it

Task	Dice $\uparrow$		HD95 (mm) $\downarrow$	
	val000	val100	val000	val100
Brats-GLI	45.9	46.6	15.6	15.1
CHAOS	85.9	86.0	22.7	20.6
ON-Harmony	66.5	66.6	4.3	4.3
Open-MS	38.5	37.9	24.3	24.2

Table 12. Applying PALETTE-Aug at validation time as well as at training time, on the four tasks where both configurations were evaluated (Mandible’s val100 arm was not evaluated on its cross-dataset test sets; the Breast, Spine and Pelvis arms were not trained). Across these tasks the two show no significant difference in either direction (same cross-task test as Tab. 1; two-sided Dice $p = 0.74$, HD95 $p = 0.096$), though val100 is significantly ahead within Brats-GLI and ON-Harmony individually. We report the like-for-like val000 configuration throughout the main paper.

Component	CHAOS	Open-MS	Brats-GLI	ON-Harmony
	T1in	FLAIR	T1n	T1w
Base (no augmentation)	36.1	25.2	26.0	27.5
+ k -means partition	84.8	30.6	41.7	67.6
+ label remap	87.7	34.6	44.3	68.0
+ Voronoi sub-parcellation	87.2	36.1	42.8	68.2
+ affine remap (PALETTE-Aug alone)	87.9	40.4	48.5	63.6
+ Auglab (Ours , deployed)	87.4	39.5	47.5	68.6

Table 13. Mechanism ablation (top five rows, additive, no Auglab) and deployment composability check (bottom row). Each rung adds exactly one PALETTE-Aug component on top of the row above. Dice (%) over all test contrasts of the setting, the training contrast included, with equal weight per contrast (the all-contrast series of the same ladders as Tab. 9, which reports the held-out contrasts only).

estly less than PALETTE-Aug and far more than the label-style methods (Tab. 14). The large margins appear against the methods that discard texture.

Method	FLAIR	T1w
PALETTE-Aug	0.809	0.894
Auglab	0.768	0.777
SynthSeg-EM	0.462	0.422
noise fill	0.457	0.401
SynthSeg-noEM	0.330	0.319

Table 14. Texture fidelity: NGF similarity to the source scan (mean over $n=30$ Open-MS scans; $\approx 1/3$ = isotropic-noise floor). The PALETTE-Aug row is the transform alone (the real-fill rung), without the Auglab pipeline.

(Sec. 4.1); Auglab is a full prior augmentation method in its own right, not a component of PALETTE-Aug, so stacking it on top of the ladder above is a composability/deployment check, not a further mechanism-ablation step. Tab. 13 keeps the two separate: the top five rows isolate PALETTE-Aug’s internal mechanism, evaluated with no Auglab throughout, while the bottom row reports PALETTE-Aug plus Auglab — the configuration used everywhere else in this paper — set apart by a rule rather than presented as the next rung of the same ladder.

12. Texture fidelity per method

The noise-versus-real NGF analysis is in the main paper (Fig. 3); Tab. 14 gives the per-method values it cites.

Why Auglab is the closest competitor. Auglab is the strongest competitor on the cross-task test, and NGF suggests why: it keeps most of the source texture, only mod-